MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, POWER, DIGITAL, KEY CODE BDH

This amendment forms a part of MIL-M-28787/119A(EC), dated 18 February 1982, and is approved for use within the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

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** 4.4.2, delete and substitute the following:

"4.4.2 Order of testing. Test condition A through W shall be performed in any order with the following exceptions. Test conditions A, B, C, E, and H are stress tests and will be performed only at 25 C ±5 C. If test conditions A, B, C, E, or H are part of a sequence of tests, the order of testing shall be as follows: E, A, B, C, and H where applicable, followed by the remaining tests in the sequence."

* 4.5.1, sixth line: Delete "through K" and substitute ", G, J, K,".

* 4.5.2, fourth line: Delete "," and add "and test condition U" between the words column and shall.

* 4.5.3, fourth line: Delete "F through K" and substitute "F, G, J, K,".

* 4.5.3, sixth line: Delete "F through W" and substitute "F, G, and J through W".

6.3, delete and substitute the following:

"6.3 Cross reference of classification. Military Specification M28787/119 supersedes Department of the Navy, Naval Ordnance System Command Drawing No. 10001-2658715. Part number M28787/119-1 supersedes part numbers 2658715 and M28787/119."

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FIGURE 5: Delete and substitute new figure 5 as printed on page 3 of this amendment.

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* TABLE V, test condition E1, under B to pin column: Delete "All other pins tied together" and substitute "All other pins tied together except pin 11".

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FIGURE 6: Delete and substitute new figure 6 as printed on page 4 of this amendment.

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FIGURE 8: Delete and substitute new figure 8 as printed on page 5 of this amendment.

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* TABLES XV, XVI, and XVII: Delete and substitute new tables XV, XVI, and XVII as printed on page 6 of this amendment.

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* FIGURE 17, note 5, test condition: Delete "S1" and substitute "T1".

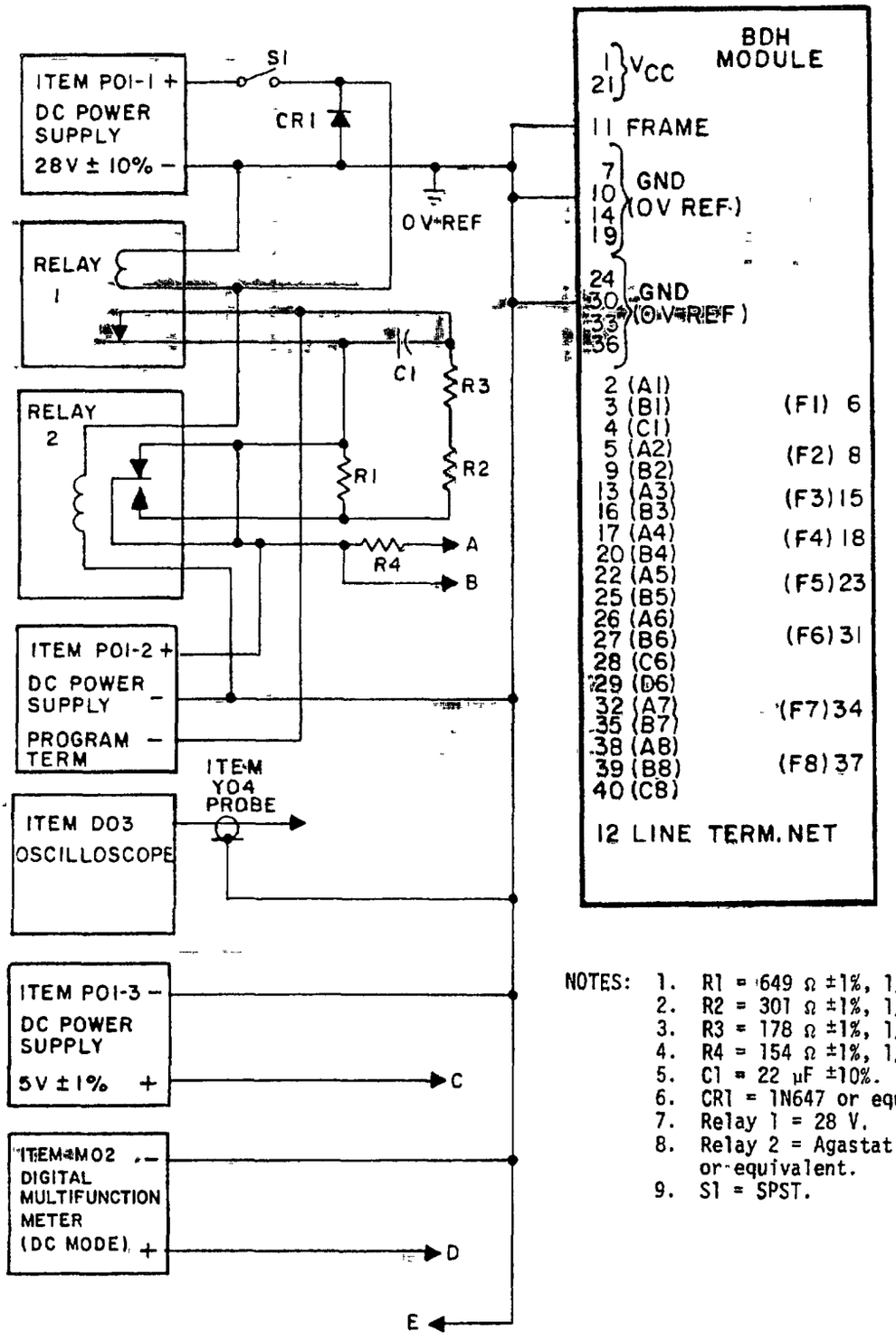
* FIGURE 17, note 6, test condition: Delete "S2" and substitute "T2".

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FIGURE 20: Delete and substitute new figure 20 as printed on page 7 of this amendment.

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

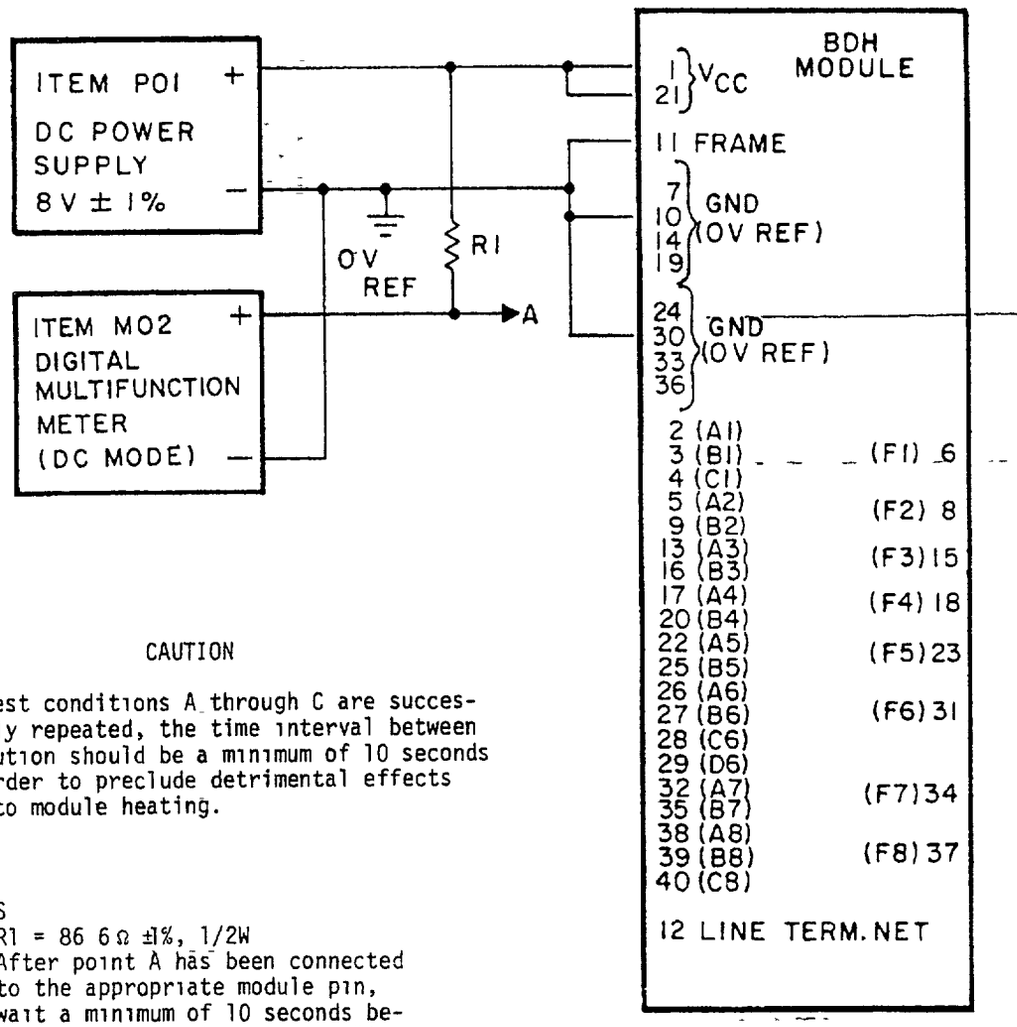
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NOTES:
1. R1 = 649 Ω ±1%, 1/8 W.
2. R2 = 301 Ω ±1%, 1/8 W.
3. R3 = 178 Ω ±1%, 1/8 W.
4. R4 = 154 Ω ±1%, 1/8 W.
5. C1 = 22 µF ±10%.
6. CR1 = 1N647 or equivalent.
7. Relay 1 = 28 V.
8. Relay 2 = Agastat Model 2112 or equivalent.
9. S1 = SPST.

FIGURE 5. Zener applied potential test circuit for table III.

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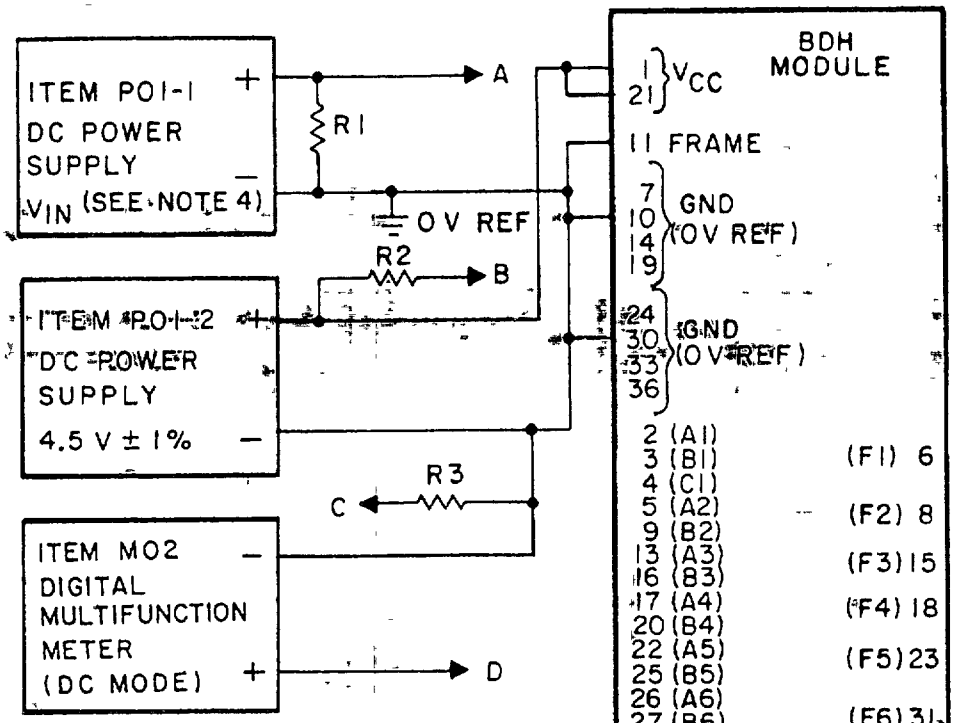
CAUTION

If test conditions A through C are successively repeated, the time interval between execution should be a minimum of 10 seconds in order to preclude detrimental effects due to module heating.

NOTES
1. R1 = 86.6 Ω ±1%, 1/2W
2. After point A has been connected to the appropriate module pin, wait a minimum of 10 seconds before measuring the output voltage.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV

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NOTES.
1. R1 = 53.6Ω ±1%, 1/2W.
2. R2 = 115Ω ±1%, 1/2W.
3. R3 = 16.9 kΩ ±1%, 1/8W.
4. Use appropriate V_{IN} for specified test condition and temperature listed below.

Test condition	Temperature (°C)	V_{IN} (volts)
F1-F20	0	1.20
	25	1.10
	60	0.95
	80	0.80
G1-G8	0	2.0
	25	1.9
	60	1.8
	80	1.8

0°, 60°, and 80°C refer to operating temperatures (module fin); 25°C refers to ambient temperature. For 80°C use the 0° to 60°C end-of-life limits.

FIGURE 8. Output voltage (high-level and low-level) test circuit for table VI.

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TABLE XV. Terminator amplitude (10 MHz and 1 MHz) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		C to pin	Min	Max		
T1	Terminator amplitude, 10 MHz	B	0.940	1.220	Vp-p	17
T2	Terminator amplitude, 1 MHz	B	1.786	1.974	Vp-p	17

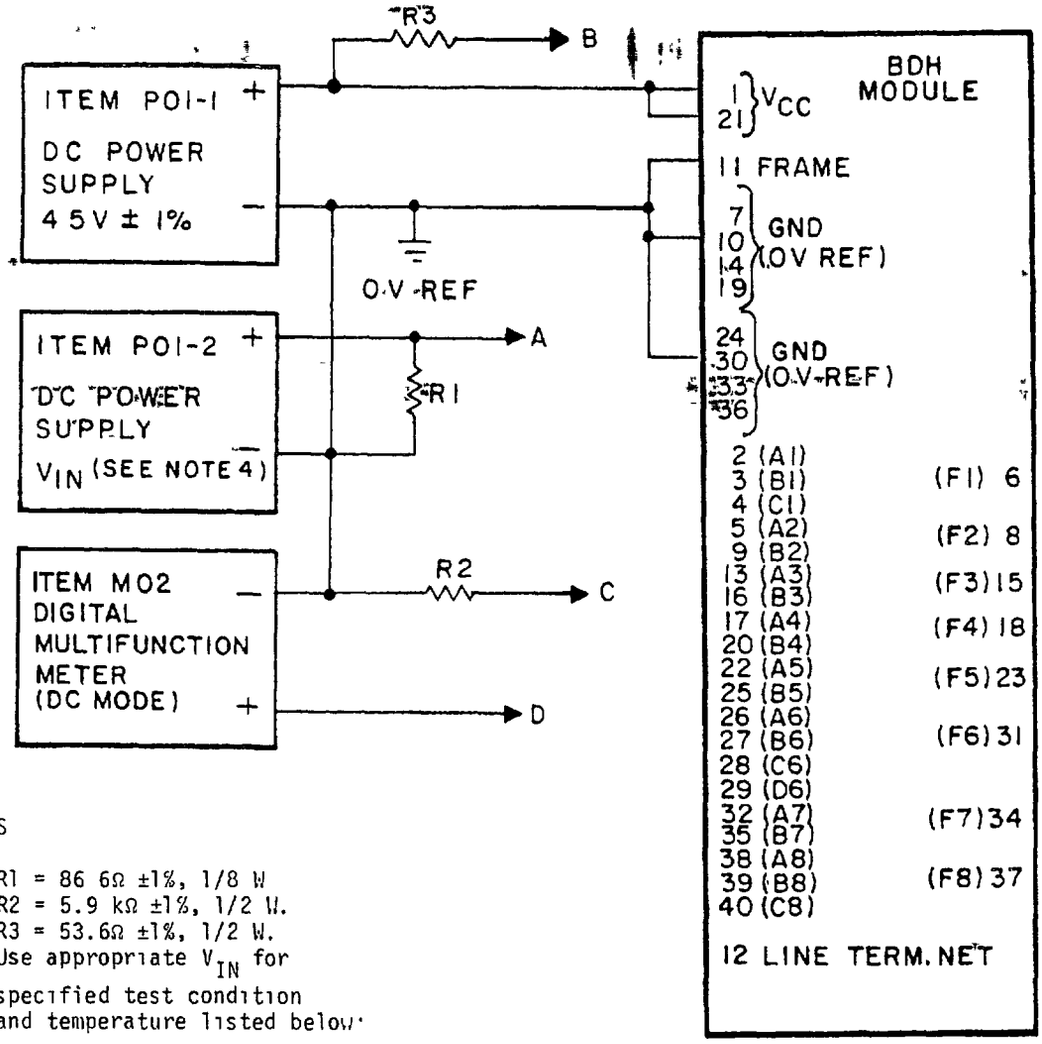
TABLE XVI. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient rise time	10	140	µs	18, 19

TABLE XVII. Parallel operation output voltage (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	Parallel operation output voltage, high level	2,3,4,5, 9,13,16	Open	6,8,15	6,8,15	3.8	5.5	V	20
V2	"	17,20,22, 25,26,27, 28,29	"	18,23,31	18,23,31	"	"	"	"
V3	"	26,27,28, 29,32,35, 38,39,40	"	31,34,37	31,34,37	"	"	"	"
W1	Parallel operation output voltage, low level	2,3,4,5, 9,13,16	6,8,15	Open	5,8,15	0.00	0.40	"	"
W2	"	17,20,22, 25,26,27, 28,29	18,23,31	"	18,23,31	"	"	"	"
W3	"	26,27,28, 29,32,35, 38,39,40	31,34,37	"	31,34,37	"	"	"	"

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NOTES

1 R1 = 86.6Ω ±1%, 1/8 W
2 R2 = 5.9 kΩ ±1%, 1/2 W.
3. R3 = 53.6Ω ±1%, 1/2 W.
4 Use appropriate V_{IN} for specified test condition and temperature listed below:

Test condition	Temperature °C 1/	V_{IN} ±1%
V1 - V3	0	1.20
	25	1.10
	60	0.95
W1 - W3	0	2.0
	25	1.9
	60	1.8

1/ The temperature of 25°C refers to ambient temperature: 0° and 60°C refer to operating temperature.

FIGURE 20. Parallel operation output voltage (high-level and low-level) test circuit for table XVII.

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